BUSINESS REQUIREMENTS DOCUMENT (BRD)

Project Name: Book Library Management System

Version: 1.0

1. PROJECT OVERVIEW

The Book Library Management System is a web/mobile application designed to manage digital and physical library operations. The system allows users to browse, read, rent, review, and manage books securely using JWT-based authentication and OTP-based onboarding.

The platform supports three roles:
1. User (Reader)
2. Author (Extended User Role)
3. Admin

The system ensures secure PDF access (view-only), inventory tracking, rental lifecycle management, and analytics monitoring.

2. BUSINESS OBJECTIVES

- Provide a centralized digital and physical book management platform.
- Enable secure online reading of books (non-downloadable PDF).
- Allow physical book rental and return tracking.
- Support user-to-author upgrade functionality.
- Provide analytical insights to administrators.
- Improve engagement via wish list, ratings, and reviews.
- Implement secure authentication using OTP and JWT.

3. SYSTEM SCOPE

IN-SCOPE FEATURES:
- OTP-based user registration (Email or Mobile).
- JWT-based authentication and role-based authorization.
- Book browsing and filtering.
- Secure PDF streaming (no download).
- Physical rental system.
- Return book functionality.

- Wishlist feature.
- Rating and review system.
- Author onboarding and approval process.
- Admin analytics dashboard.
- Automated email notifications.

OUT-OF-SCOPE (Future Enhancements):
- Payment gateway integration.
- Mobile native app.
- Third-party distribution systems.

4. USER ROLES AND RESPONSIBILITIES

4.1 USER (READER)
- Register using email or mobile number.
- Verify account via OTP.
- Login securely (JWT token generated).
- Browse and search books.
- Filter by genre, rating, popularity.
- Read books online (view-only PDF).
- Rent physical books.
- Click 'Return Book' button to initiate return.
- Add/remove books from wishlist.
- Provide ratings (1-5 stars).
- Submit reviews.
- View dashboard (Books read, Active rentals, Rental history, Wishlist, Reviews).

4.2 AUTHOR (EXTENDED USER ROLE)
- Submit biography, qualifications, experience, photo, manuscript.
- Admin approval required.
- Submit new books.
- Upload manuscripts.
- Edit before approval.
- Track submission status.
- View performance metrics (reads, rentals, ratings, reviews).

4.3 ADMIN
- Approve/reject authors.
- Add/edit/delete books.
- Approve/reject submissions.
- Manage inventory.
- Moderate reviews.
- View user and author data.

- Access analytics dashboard.
- Mark dispatch.
- Monitor rentals and returns.

5. AUTHENTICATION & SECURITY REQUIREMENTS

- OTP-based onboarding.
- JWT authentication.
- Role-based access control.
- Encrypted password storage.
- Token expiration management.
- Secure backend PDF streaming endpoint.

6. BOOK MANAGEMENT REQUIREMENTS

Each book must include:
- Title
- Author details
- Genre
- Description
- ISBN
- Publication date
- Cover image
- Secured PDF
- Total copies
- Available copies
- Ratings (average)
- Review count
- Wishlist count
- Read count
- Rental count

7. SECURE PDF VIEWING REQUIREMENTS

- PDF streamed from the backend.
- No direct file access.
- Download disabled.
- Authenticated access only.
- Token validation is required.

8. RENTAL MANAGEMENT REQUIREMENTS

Rental Lifecycle:
1. Rental Request

2. Rental Confirmation Email
3. Admin marks as Dispatched
4. Dispatch Email sent
5. Due Date Tracking
6. User clicks 'Return Book'
7. Return Confirmation Email
8. Admin marks as completed

9. RETURN BOOK FEATURE

When user clicks 'Return Book':
- Status = Return Initiated
- Admin notified
- User confirmation email
- Admin confirms physical receipt
- Inventory updates
- Rental marked Completed

10. WISHLIST FEATURE

- Add/remove wishlist
- Track wishlist count
- Admin can view most wishlisted books

11. RATING & REVIEW SYSTEM

- 1–5 star rating
- Written feedback
- Average rating calculation
- Admin moderation

12. ANALYTICS DASHBOARD

System Overview:
- Total Users
- Total Authors
- Total Books
- Active Rentals
- Overdue Rentals

Book Analytics:
- Most Read
- Most Rented
- Most Wishlisted

- Highest Rated

Author Insights:
- Top Performing Authors
- Books per Author
- Reads per Author
- Rentals per Author

13. NON-FUNCTIONAL REQUIREMENTS

- Responsive UI
- Secure transmission
- Scalable architecture
- Reliable email system
- OTP expiration
- Fast search performance

14. SUCCESS CRITERIA

- OTP onboarding works
- JWT authentication works
- RBAC enforced
- Secure PDF streaming
- Rental & return lifecycle works
- Email notifications triggered
- Analytics operational
- Inventory auto-updates

15. DEVELOPMENT PROCESS REQUIREMENT

Students must follow this order:
Step 1: Finalize Technology Stack
Step 2: Design Database (ER Diagram + Schema)
Step 3: Backend Development
Step 4: API Documentation (Swagger/OpenAPI)
Step 5: Frontend Development
Step 6: Testing & Submission
Step 7 (Optional Bonus): Docker Deployment

16. UI DEVELOPMENT CONSTRAINT

Students MUST NOT use:
- Bootstrap
- Tailwind

- Material UI
- Any pre-built CSS libraries

Students must build UI using:
- Custom CSS
- CSS Modules
- SCSS

UI must be responsive and professionally designed.

17. API DOCUMENTATION REQUIREMENT

Students must:
- Implement Swagger/OpenAPI inside backend

OR
- Provide structured API documentation

Also mandatory:
- Submit exported Postman collection (.json)
- Proper grouping of APIs (Auth, Books, Rentals, Admin)

18. SOLID PRINCIPLES (MANDATORY IMPLEMENTATION)

Students must learn and apply SOLID principles:

S - Single Responsibility Principle
O - Open/Closed Principle
L - Liskov Substitution Principle
I - Interface Segregation Principle
D - Dependency Inversion Principle

Implementation expected in:
- Service layer design
- Repository pattern
- Authentication modules
- API client abstraction
- Clean separation of concerns

19. OPTIONAL BONUS

Bonus marks for:
- Dockerizing application
- docker-compose setup
- Deployment documentation

20. TECHNOLOGY STACK

- Frontend
  - React
  - Next.js
  - Angular
  - React Native (app)
  - Flutter (app)
- Backend
  - Django + Django REST Framework
  - NodeJs
  - NestJs
- Database
  - PostgreSQL